Final Project

MAT2007 Introduction to Programming

Testosterone levels as an indication of PCOS in women

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Introduction

This project details the female hormonal disorder Polycystic Ovary Syndrome (PCOS). Affecting women of reproductive age, the disorder is characterised by hyperandrogenism, which refers to an excess secretion of male sex hormones like testosterone, irregular or absent ovulation associated with the lack of monthly menstruation, and presenting polycystic ovaries.

Recently, this condition has been renamed to Polyendocrine Metabolic Ovarian Syndrome with the intention to increase patient well-being [1]. Not all PCOS-affected people qualify for all 3 characteristics, which are commonly known as the Rotterdam criteria [2]. For this reason, this project aims to investigate the statistical significance of testosterone levels to diagnose PCOS. It is hypothesised that diagnosed patients will present a statistically significant elevation in serum testosterone levels.

Dataset

The data for this analysis were accessed on Mendeley Data and contain data collected from patients with and without PCOS [3]. It provides data on serum testosterone levels as well as comprehensive patient data.

In this case study, blood sample data are analysed to explore how strongly PCOS-positive patients' (n=35) testosterone levels differ from those of normally menstruating women without PCOS (n=37). The results will provide information on average testosterone levels to be expected in PCOS patients as well as prove the use of blood samples to aid diagnostics. The concrete research question is "To what extent are serum testosterone levels significantly elevated in women with Polyendocrine Metabolic Ovarian Syndrome (PMOS) compared to a healthy control group?"

Methods

The data analysis was conducted using Python. The data analysis was conducted on the raw data read from the Excel workbook, using entries from the serum testosterone measurements. To explore the average of both the CON (control) and PCOS groups, a bar plot was visualised that combines mean bars with an overlapping strip plot of raw data points. Additionally, dual error bars of a thin outer line representing the ± 1 standard deviation (SD) and a thick line representing the ± 1 standard error (SE) were included.

Moreover, a kernel density estimation is used to show the probability density of both groups with semi-transparent fills to show the overlapping distributions.

The last figure integrates the kernel density shapes in a violin plot, essentially flipping the curve by 90 degrees as well as adding the individual data points to show the entire measurement range.

Finally, to evaluate whether the observable difference in the plots translates into a numerical statistical significance, the script includes arithmetic analysis of the dataset's groups as two distinct, non-identical (independent) groups of individuals. [4].

The Welch's t-test was conducted, and the analysis calculated the sample size n, arithmetic means $\bar{X}$, sample standard deviations, and standard errors. Furthermore, it applied the Welch-Satterthwaite equation to calculate the 95% confidence intervals.

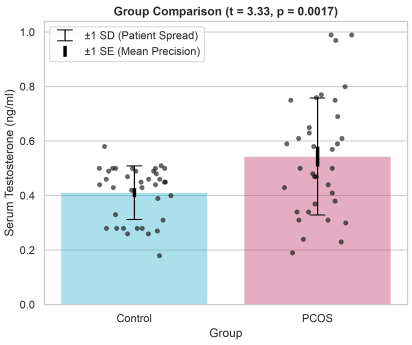
Results and Discussion

Table 1: Results of Statistical Operations

Group	n	Mean (ng/mL)	SD	SE	p-value
Control	37	0.4108	0.0985	0.0162	0.0010
PCOS	35	0.5431	0.2148	0.0363	0.2465

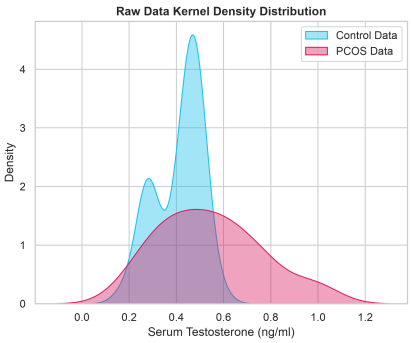
95% Confidence interval of the difference = [0.0524, 0.2123], t = 3.33, p = 0.0017.

Plot 1: Group Comparison



Discussion 1:
In this plot, which shows the group means for serum testosterone, it is clear that the control group and PCOS patients show a difference. It incorporates dual error bars indicating the SD and SE. The combined bar and strip plot highlights both mean differences and the spread of individual observations. The wide SD for PCOS indicates biological heterogeneity within a cohort. The addition of the jittered strip plot also shows how tightly the raw data points are clustered. As expected, in unaffected patients the spread is smaller. The narrow SE for both groups indicates a high precision in estimating the true group means, allowing for more meaningful conclusions.

Plot 2: Kernel Density



Discussion 2:
The kernel density estimation shows the probability density curves of both groups. The semi-transparent fills show the overlapping distributions and clearly demonstrate the elevated probability density of testosterone levels in PCOS patients compared to the control group. The control group has the highest probability around 0.48 ng/mL with a noticeable dip around 0.35 ng/mL. Due to the small sample size (n=37), the shape of this curve is sensitive to the smoothing technique used to generate it. Consequently, rather than being a characteristic of the population, the drop may be an artefact of the sampling. Nevertheless, neither of the groups demonstrates a severe enough skewness to affect the t-test.

Plot 3: Violin Plot



Discussion 3:
Figure 3 combines the density shapes to show the full distribution pattern with the individual data points from the strip-plot. This plot was added for easier comparison and can have to have a visual representation of the distribution and clustering of individual data points. The broader body of both the groups' violins emphasises that there is a large density at intermediate testosterone levels. The height of the PCOS violin, however, represents that larger range of testosterone levels, whereas healthy patients cluster around their respective mean.

To determine if the difference between the two groups is statistically significant, the Welch's t-statistic was calculated using the sample means ($\bar{X}$), standard deviations (SD), and sample sizes (n) [5]:

$$t = \frac{\bar{X}_{PCOS} - \bar{X}_{CON}}{\sqrt{\frac{SD^2_{PCOS}}{n_{PCOS}} + \frac{SD^2_{CON}}{n_{CON}}}} \tag{1}$$

This gave a t-statistic of 3.3288. Moreover, a p-value of 0.0017 was obtained. Because $p < 0.05$, the elevated testosterone levels in the PCOS group are statistically significant.

Conclusion

The serum testosterone levels of PCOS patients are statistically significantly elevated. Higher average testosterone values however, does not make this an ideal diagnostic tool. Still, there is an area of overlap that shows that even a positive diagnosis can manifest with normal testosterone levels. Thus, testosterone levels alone are not the sole determining factor for a diagnosis.

References

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